# Campus Safety & Incident Analytics System

## Logical Database Design & Relational Schema Report

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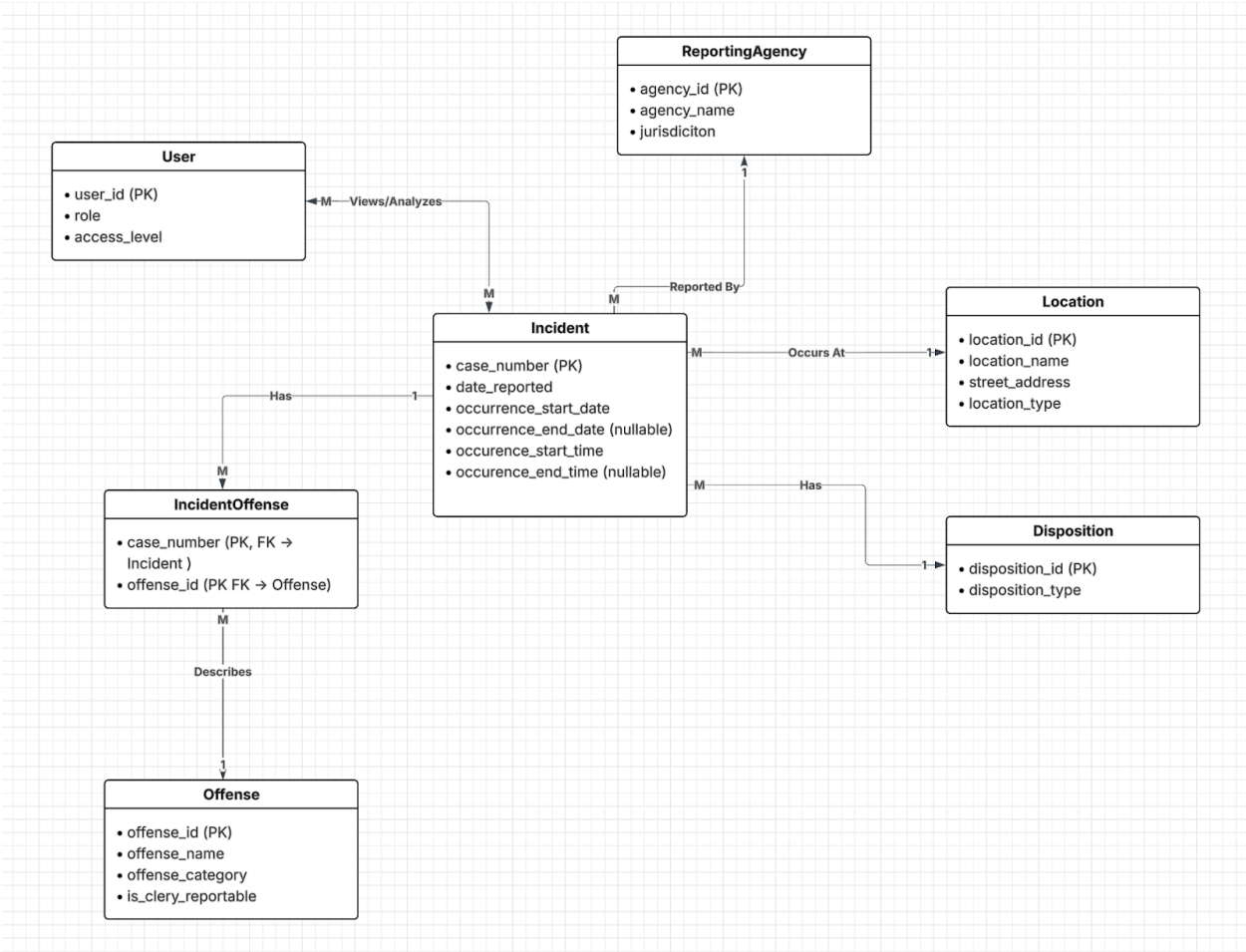
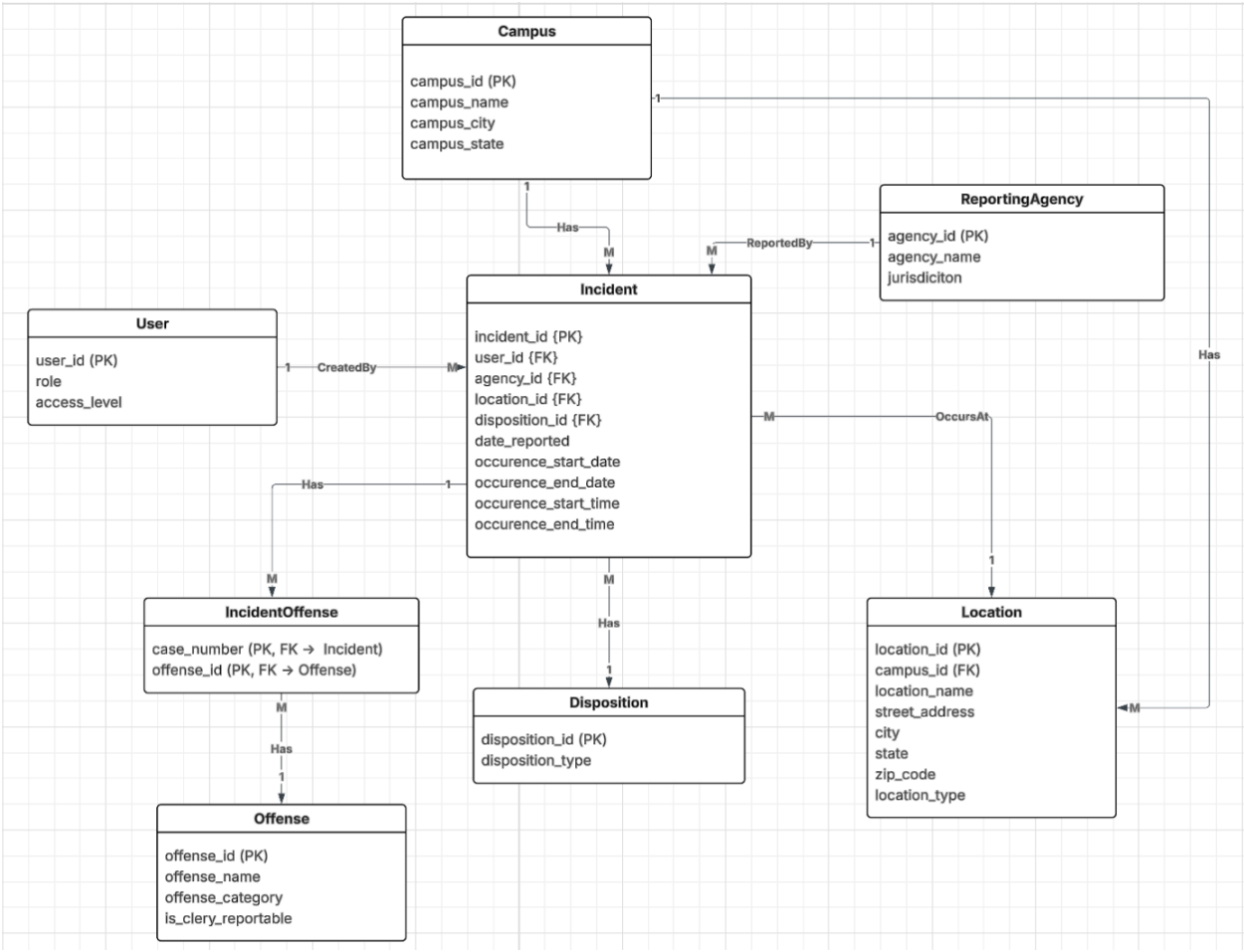
# 1.Conceptual ER Diagram



# 2. Logical ER Diagram

This section presents the transformation of the conceptual ER model into a logical database design. The logical ER diagram incorporates primary keys, foreign keys, and refined relationships to ensure compatibility with relational database implementation.

## 2.1. Logical ER Diagram Overview




## 2.2. Design Changes and Justifications

1. Addition of CreatedBy Relationship (User–Incident)

The relationship between User and Incident was refined by introducing a one-to-many (1:M) “CreatedBy” relationship. In the conceptual design, the association between users and incidents was implicit. In the logical design, this relationship is implemented by adding user_id as a foreign key in the Incident entity. This ensures that each incident is associated with exactly one user, while allowing a user to create multiple incident records. This follows standard relational mapping principles for one-to-many relationships, where the primary key of the “one” side is

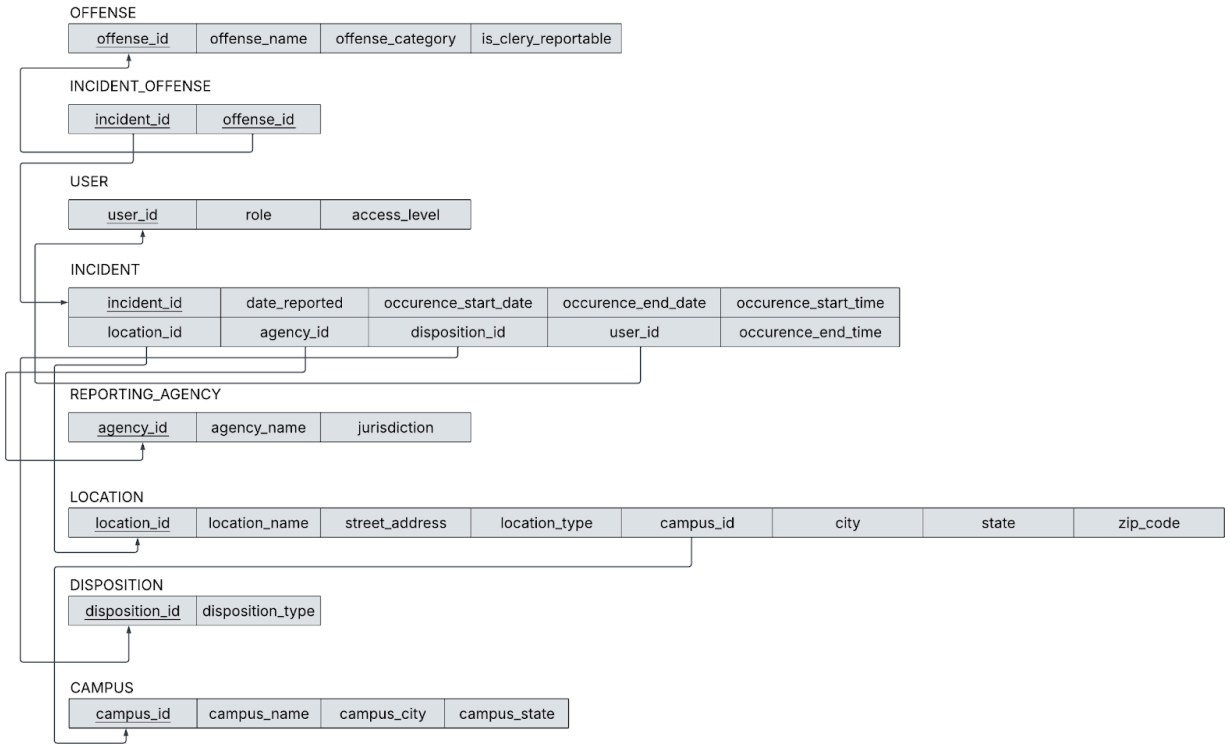
included as a foreign key in the “many” side. This refinement improves data integrity and accountability by ensuring all incident records are associated with a known user.

2. Enhancement of the Location Entity (Addition of Address Attributes)

The Location entity was expanded to include additional attributes: city, state, and zip_code. In the conceptual design, location data was stored in a more general format. Separating address components into individual attributes ensures atomicity and improves query flexibility. This allows filtering and analysis by geographic attributes and supports future scalability. This change aligns with normalization principles by avoiding composite or multi-valued attributes.

3. Refinement of Primary Key Definitions

Primary keys were explicitly defined for all entities in the logical design. Surrogate keys such as incident_id, user_id, and location_id were introduced to ensure uniqueness and avoid reliance on natural keys, which may be inconsistent or subject to change. This improves indexing performance, enforces entity integrity, and ensures reliable referencing through foreign key relationships in the relational schema.

# 3. Relational Schema Diagram

This diagram represents the relational schema derived from the logical ER design. Each entity has been converted into a table, with primary keys and foreign keys clearly defined. The schema enforces referential integrity and supports efficient querying of incident data across related tables. This schema was implemented in a MySQL database and integrated into a Flask-based web application for querying and analysis.